Heathrow Community Noise Monitoring Analysis: Stanwell Moor

Summary of key findings

A community noise monitor was deployed by Heathrow Airport in Stanwell Moor, Spelthorne between 14 May 2025 and 27 November 2025 inclusive.

This report provides an analysis of aircraft flight tracks and noise measurements recorded over the 198-day monitoring period. The key findings are as follows:

- Over the monitoring period, 74% of Heathrow operations were westerly, with easterly operations occurring for the remaining 26% of the time.
- During westerly operations the monitor location is overflown by departures from the southern runway (27L) at an average height of approximately 1,300 ft.
- During easterly operations the monitor location is overflown by aircraft on approach to the southern runway (09R) at an average height of 330 ft.
- During the monitoring period approximately 62,000 aircraft noise events were recorded by the Stanwell Moor monitor. The majority of noise events were from westerly departures on the southern runway (27L).
- Narrow-body A320neo aircraft were responsible for the largest share of departure noise events, representing 21% of all departure events, followed by the original A320 variant and the wide-body B777, accounting for 16% and 14%, respectively.
- The average measured departure L_{ASmax} across all aircraft types was 82.7 dB. The noisiest aircraft on average was the A380, with a measured departure L_{ASmax} of 91.0 dB.
- The quietest aircraft on departure was the narrow-body A220, followed by the A320neo, with average L_{ASmax} levels of 75.9 dB and 77.1 dB, respectively. Average departure noise levels for the re-engined A320neo and A321neo aircraft were approximately 3 dB to 4 dB quieter relative to their older A320 and A321 counterparts.
- The average measured arrival L_{ASmax} across all aircraft types was 89.2 dB. The noisiest aircraft on arrival was the B767, followed by the A380, with average L_{ASmax} levels of 92.1 dB and 91.6 dB, respectively.
- The quietest aircraft on arrival was the A220, followed by the A320neo, with average L_{ASmax} levels of 85.4 dB and 85.8 dB, respectively. The average arrival noise level for the A320neo was approximately 1 dB quieter relative to its older A320 counterpart.

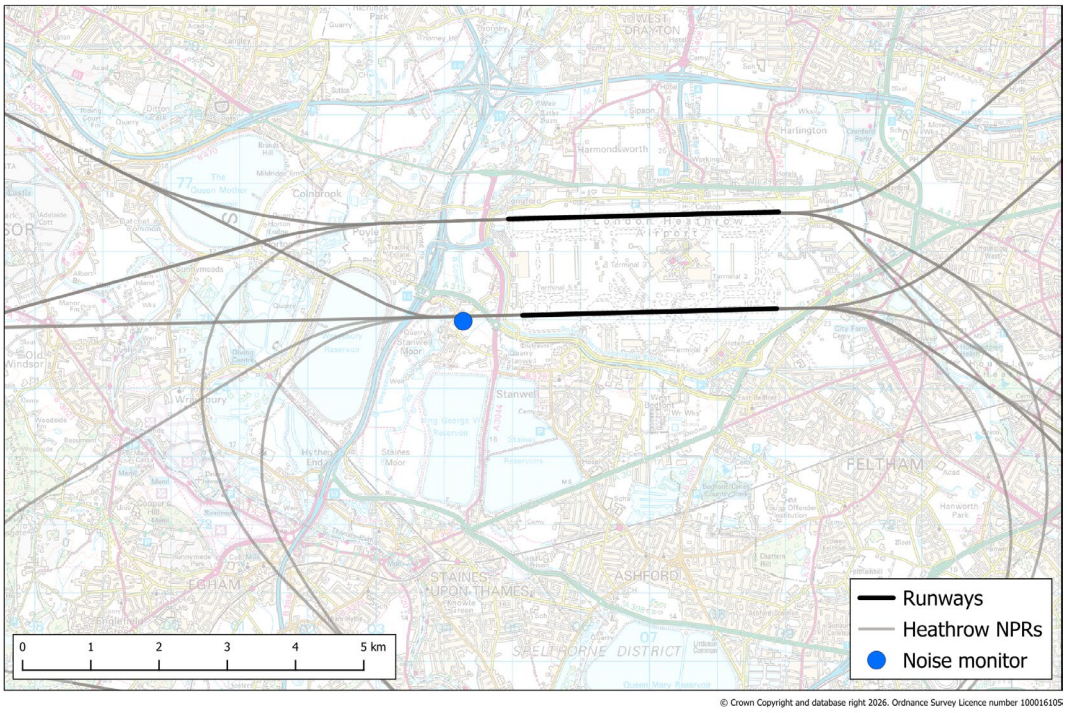
Introduction

1. A community noise monitor was deployed by Heathrow Airport in Stanwell Moor, Spelthorne between 14 May 2025 and 27 November 2025 inclusive.

2. The northern edge of Stanwell Moor is situated directly under the initial westerly departure Noise Preferential Routes (NPRs) from Heathrow's southern runway, approximately 0.5 km from the airfield boundary. During easterly operations the same area is overflown by arrivals to the southern runway.

3. Figure 1 shows the position of the noise monitor, which was located in the rear garden of a property on Horton Road, approximately 300 m from the nearest main road (A3113 Airport Way) and 600 m from the M25 motorway.

Figure 1 Stanwell Moor noise monitoring site



4. This report by the Environmental Research and Consultancy Department (ERCD) of the Civil Aviation Authority (CAA) was commissioned by Heathrow Airport to provide an analysis of aircraft flight tracks and noise measurements recorded over the 198-day monitoring period at Stanwell Moor.

5. Explanations of technical terms used in this report can be found in Appendix A.

Heathrow operations

Background

6. In general, aircraft will take off and land into the wind to maximise lift for safety reasons. The wind direction, which varies over the course of a year, will therefore have an important influence on the usage of runways. During the daytime a 'westerly preference' is operated at Heathrow, which means that during periods of light easterly winds aircraft will continue to take off and land in a westerly direction.

7. Westerly operations at Heathrow typically occur for about 70% of the time, with easterly operations occurring for the remaining 30% of the time. Shorter term fluctuations between westerly and easterly operations can vary considerably from long-term averages.

8. During westerly operations, runway alternation is applied during the daytime on a weekly schedule. This provides for one runway to be used for landings and the other for take-offs between 06:00-15:00. At 15:00 the use of the runways is then switched until the last departure of the day. After the last departure of the day, a four-week night-time runway alternation pattern is also utilised until 06:00 the following morning.[1]

9. There is currently no runway alternation during the day on easterly operations due to the legacy of the Cranford Agreement, which prohibited easterly departures from Heathrow's northern runway other than in limited circumstances. During easterly operations, the majority of departures use the southern runway and the majority of arrivals use the northern runway.

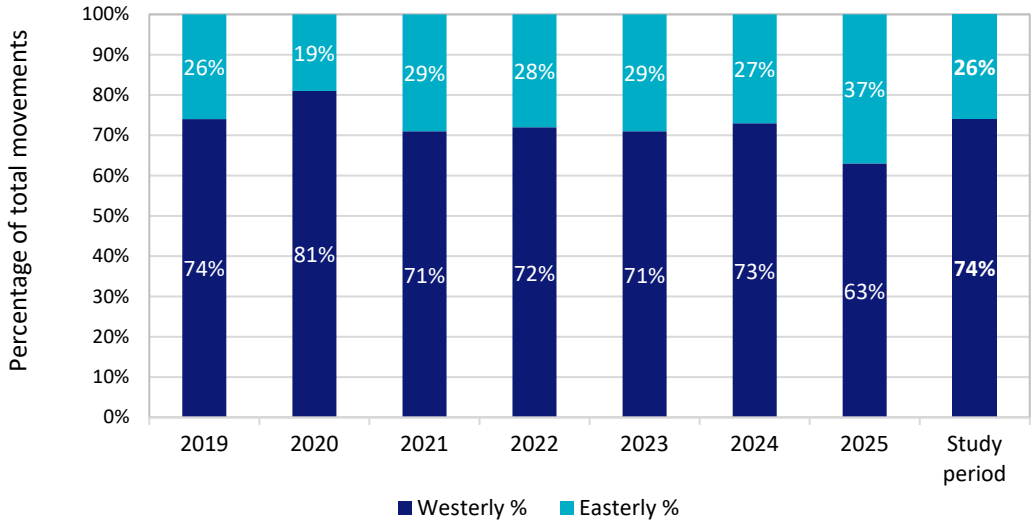
[1] Runway alternation | Heathrow

Operations during the monitoring period

10. During the monitoring period, between 14 May 2025 and 27 November 2025 inclusive, 74% of operations were westerly, with easterly operations occurring for the remaining 26% of the time.

11. For context, Figure 2 compares the percentage of easterly and westerly operations during the study period with annual data since 2019, based on flights recorded in Heathrow's Noise and Track Keeping (NTK) system.

Figure 2 Breakdown of easterly and westerly operations at Heathrow by year



12. Figures 3 and 4 present flight tracks for typical days of westerly departures and easterly arrivals at Heathrow, respectively, in relation to the Stanwell Moor noise monitor. During westerly operations the monitor location is overflown by departures from the southern runway (27L). The average height of westerly departures over the monitor location during the study period was approximately 1,300 ft.

13. During easterly operations the monitor location is overflown by arriving aircraft on a 3-degree approach to the southern runway (09R). The average height of easterly arrivals over the monitor location during the study period was 330 ft.

14. While the average height of easterly arrivals over the monitor location is significantly lower than for westerly departures, it should be remembered that the majority of easterly arrivals use the northern runway (09L) due to the legacy of the Cranford Agreement, see paragraph 9. This means that, combined with the overall tendency for westerly operations, Stanwell Moor experiences significantly more overflight from departures compared to arrivals.

Figure 3 Flight tracks for a typical day of westerly departures

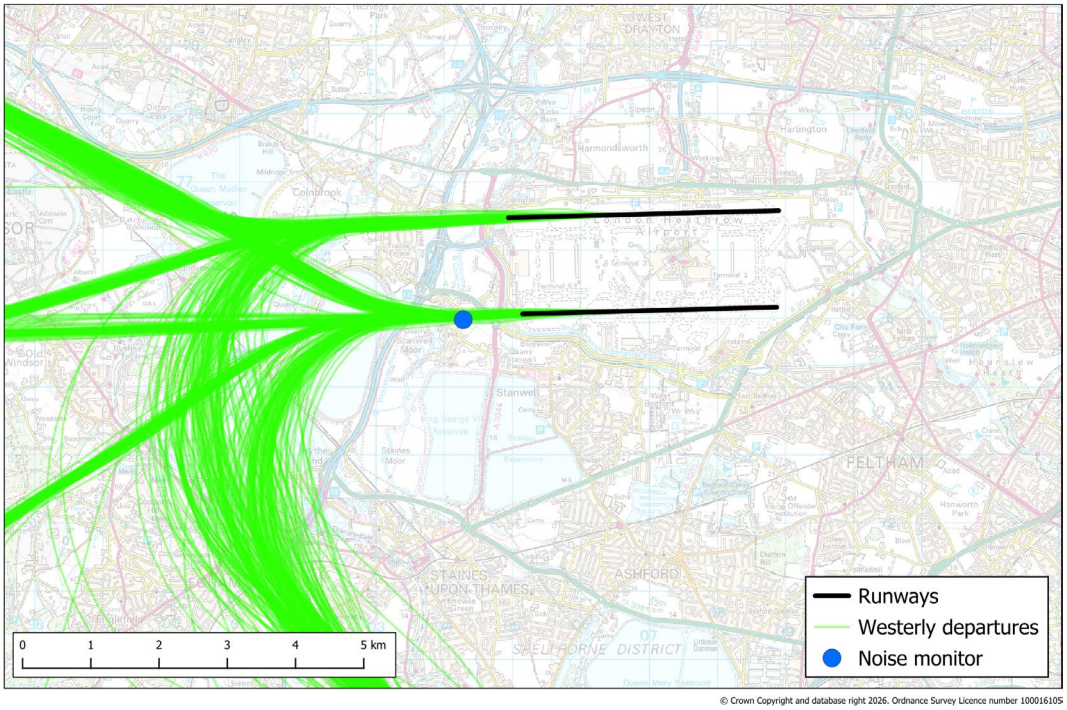
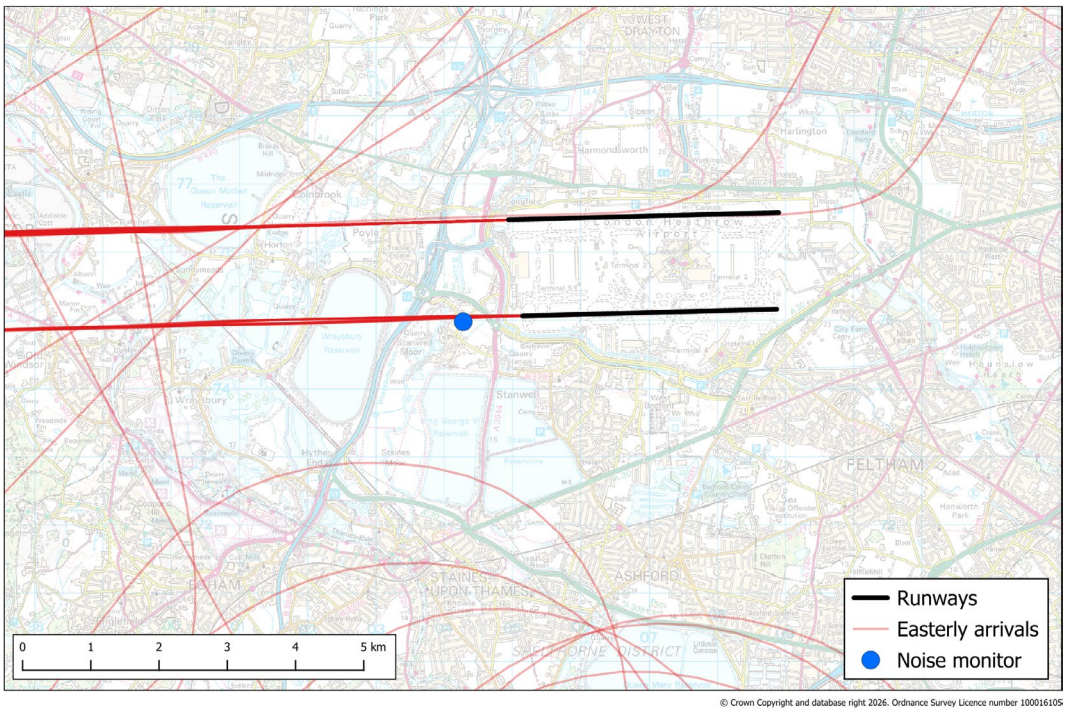


Figure 4 Flight tracks for a typical day of easterly arrivals



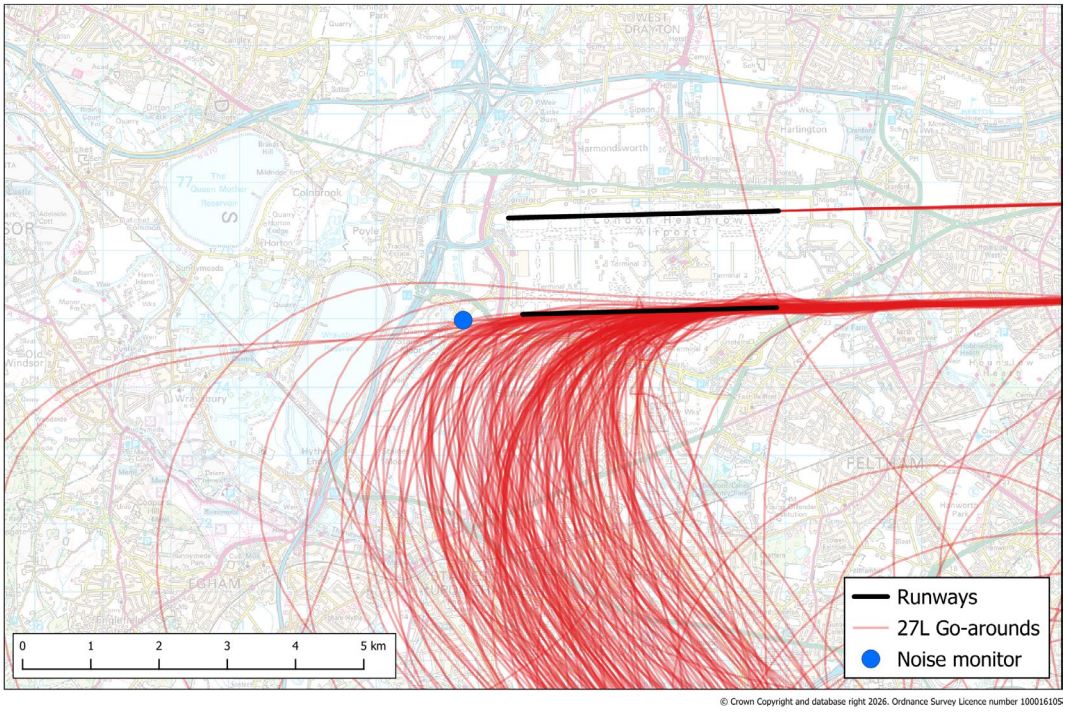
Overflight from runway 27L go-arounds

15. In addition to being overflown by westerly departures and easterly arrivals, Stanwell Moor also experiences occasional overflight by aircraft on the westerly approach to the southern runway (27L) performing a go-around procedure for safety reasons.

16. The procedure for a missed westerly approach to the southern runway requires aircraft to turn in a southerly direction following an aborted landing, see Figure 5. During a typical go-around, engine power is increased to provide maximum available thrust and the aircraft then climbs to an altitude of 3,000 ft before circling back to attempt another landing (typically to the same runway but occasionally to the opposite runway).

17. Between 14 May 2025 and 27 November 2025 there were 235 westerly arrivals to the southern runway that performed a go-around procedure, representing about 0.5% of all approaches to the southern runway over the same period. While the majority of these go-arounds turned to the south before reaching Stanwell Moor, the average height of those that overflew the noise monitor was approximately 2,000 ft.[2]

Figure 5 Flight tracks for runway 27L go-arounds during the study period



[2] The average height was calculated based on overflights with a minimum elevation angle of 48.5 degrees relative to the noise monitor.

Analysis of noise monitoring results

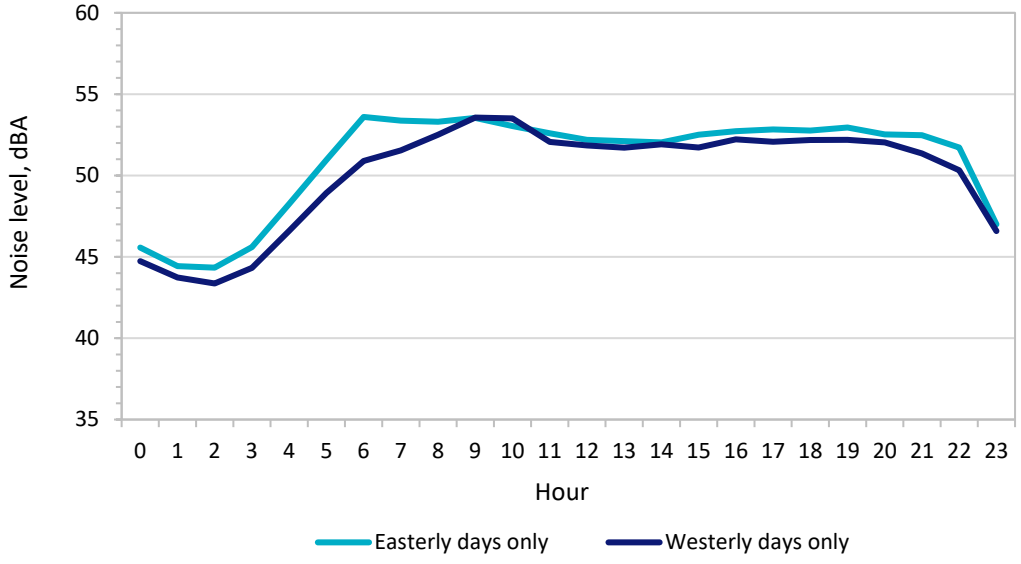
Background noise

18. The background noise at any location is usually a combination of many sources including local road traffic, distant motorways, railway lines, farm machinery and industrial equipment. The background noise level can also be affected by natural events, including weather events and birdsong. While noise monitors are equipped with windshields, the noise generated by trees and grass during windy conditions can also affect the measured noise level.

19. Figure 6 shows the average hourly background noise level ($L_{A90,1hr}$) recorded by the Stanwell Moor noise monitor over the 24-hour period, separated by direction of operation, easterly or westerly.

20. As can be seen, while background noise levels are broadly comparable for each operating direction, in general, slightly higher background levels were recorded during periods of easterly operation. This is when the prevailing wind direction (from the east) would typically place the site downwind of the airfield and the nearby A3044 and A3113 main roads (see Figure 1), increasing noise at the site compared to when the prevailing wind is from the west and the site is upwind of these noise sources.

Figure 6 Average hourly background ($L_{A90,1hr}$) noise levels



21. The overall pattern in Figure 6 is largely in line with expected results; during the early hours of the morning, the average hourly background noise level is typically at its lowest, rising above 50 dBA after 06:00 for the rest of the day until the late evening, before falling towards the start of the night period. This pattern broadly coincides with the normal increase in road traffic levels during the daytime.

Aircraft noise events

22. A noise event is recorded in the Heathrow NTK system whenever the measured noise level at a monitor exceeds a specified threshold level for longer than a minimum duration.

23. The Stanwell Moor noise monitor was set up to record noise events above a pre-determined threshold level of 64 dBA for a duration of at least 5 seconds. Any aircraft noise events registering a noise level below this threshold or exceeding the threshold but for less than 5 seconds were not recorded. While the threshold level of 64 dBA was sufficiently low enough to capture noise events for aircraft operating on the southern runway, it is important to note that some aircraft operating on the northern runway (1.4 km to the north) may have been audible at the monitor location but not noisy enough to trigger the noise event detection threshold.

24. During the monitoring period approximately 62,000 aircraft noise events in total were recorded by the Stanwell Moor monitor. Approximately 10% of the recorded noise measurements were subsequently rejected for further analysis. The majority of these were rejected due to unfavourable weather conditions (wind speeds greater than 10 m/s or during periods of precipitation) in accordance with recommended international guidance on aircraft noise monitoring.[3] A small number of events were also rejected due to evidence of contamination from non-aircraft noise sources.

25. Table 1 summarises the number of remaining Heathrow noise events that were considered valid for further analysis, separated by operation type and runway. This shows that westerly departures on the southern runway (27L) accounted for the majority of valid noise events (81%), which is unsurprising given the relative location of the monitoring site and the overall tendency for westerly operations.

Table 1 Counts of Heathrow noise events at the Stanwell Moor monitor by operation and runway (after filtering for adverse weather conditions)

	09L	09R	27L	27R	Total
Arrivals	111	1,464	37	7	1,619
Departures	0	0	43,868	10,304	54,172

[3] ISO 20906:2009, Acoustics - Unattended monitoring of aircraft sound in the vicinity of airports

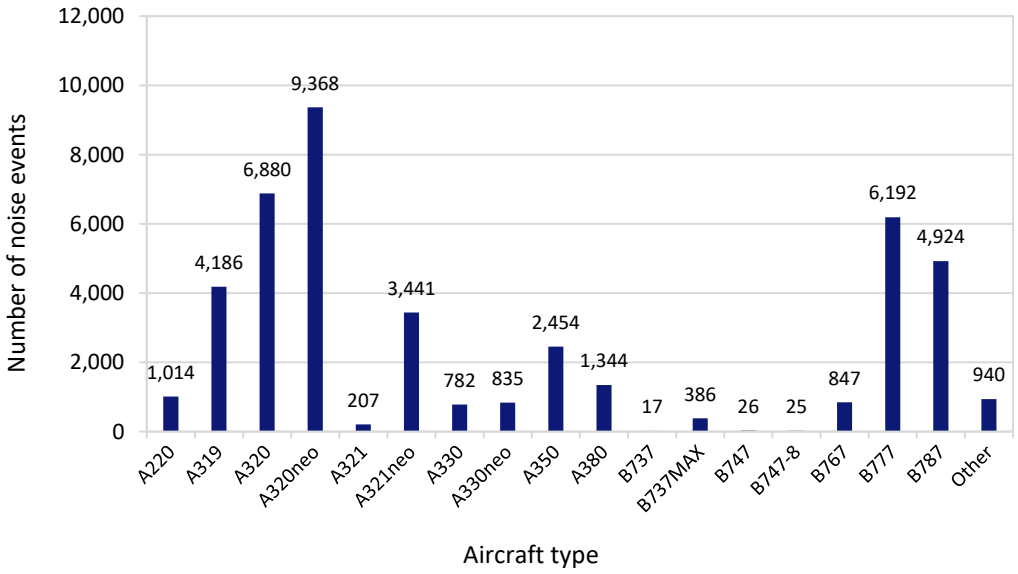
26. For arrivals, the results show that the majority of valid events were recorded for aircraft landing on the southern runway (09R). Again, this is unsurprising given the greater noise propagation distance to the monitor location for arrivals on the northern runway.[4] A small number of noise events were also recorded for westerly arrivals to runways 27L and 27R, which were related to aircraft performing go-around procedures for safety reasons (see paragraph 15).

27. Given the location of the Stanwell Moor noise monitor relative to Heathrow's southern runway, the subsequent analysis primarily focuses on noise from westerly 27L departures and easterly 09R arrivals.

Runway 27L departure noise

28. Figure 7 illustrates the total number of valid 27L departure noise events by aircraft type recorded at the Stanwell Moor monitor. The results show that the narrow-body A320neo aircraft are responsible for the largest share of those noise events, representing 21% of all departure events, followed by the original A320 variant and the wide-body B777, accounting for 16% and 14%, respectively.[5]

Figure 7 Number of 27L departure noise events by aircraft type



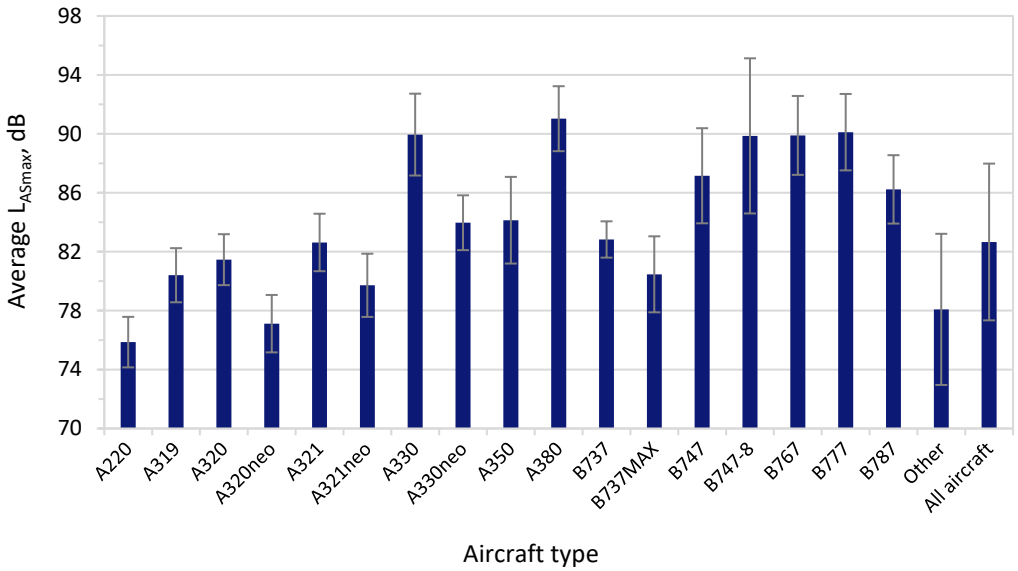
29. Figure 8 compares the average measured 27L departure L_{ASmax} noise levels by aircraft type. Error bars representing one Standard Deviation are also shown to illustrate the variation of measured levels around each average value. While the average measured departure levels for the B737, B747 and B747-8 are included in

[4] Lateral attenuation at the monitor location, which is the excess sound attenuation caused by the ground surface at lower angles of elevation, will also be significant for 09L arrivals compared to 09R arrivals.

[5] The Airbus A320, including the newer and quieter A320neo variant, is the most common aircraft type at Heathrow, accounting for more than one-third of all movements in recent years.

Figure 8 for information, caution should be applied when interpreting the results for these aircraft due to their relatively small sample sizes (see Figure 7).

Figure 8 Average measured 27L departure L_{ASmax} levels by aircraft type



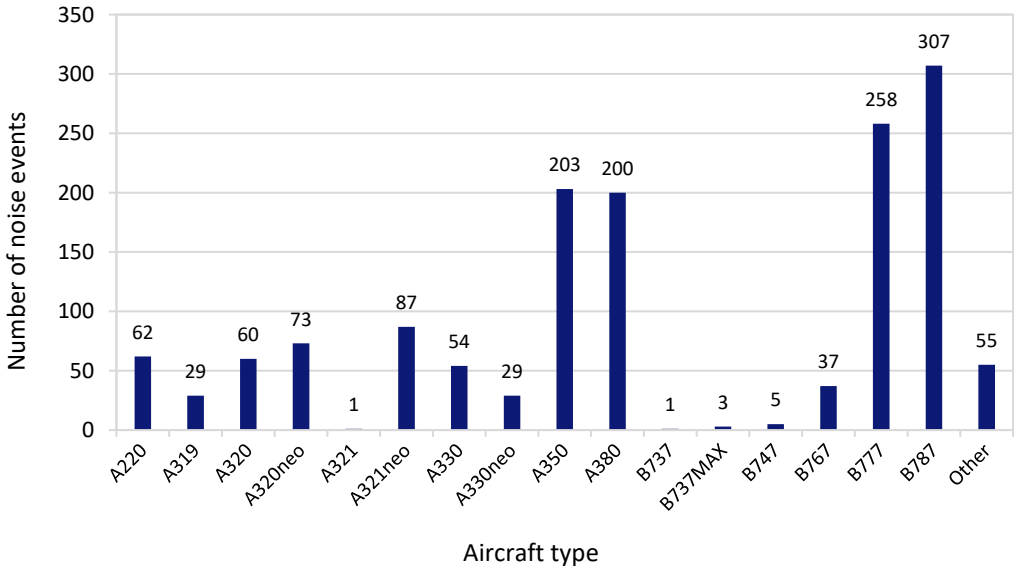
30. The average measured departure L_{ASmax} across all aircraft types was 82.7 dB. The noisiest aircraft on average was the A380, with a measured departure L_{ASmax} of 91.0 dB. However the average measured departure L_{ASmax} levels for several other wide-body aircraft all fall within approximately 1 dB of the A380 result.

31. The quietest aircraft on departure was the narrow-body A220, followed by the A320neo, with average L_{ASmax} levels of 75.9 dB and 77.1 dB, respectively. The results also show that the re-engined A320neo and A321neo aircraft are approximately 3 dB to 4 dB quieter on departure at the monitor location relative to their older A320 and A321 counterparts.

Runway 09R arrival noise

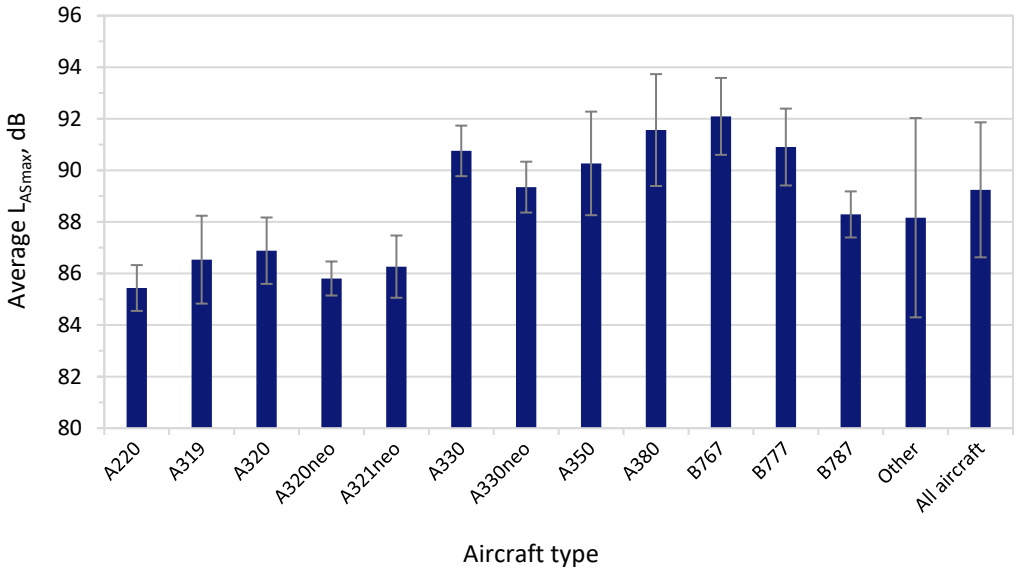
32. Figure 9 illustrates the total number of valid 09R arrival noise events by aircraft type recorded at the Stanwell Moor monitor. The results show that wide-body aircraft such as the A350, A380, B777 and B787 are responsible for the largest share of these noise events, a significant proportion of which occurred during the early morning period. This is due to the influence of Heathrow's night-time runway alternation schedule, which results in a greater tendency for early morning arrivals (which are primarily comprised of wide-body long-haul aircraft) to use the southern runway during easterly operations (see paragraphs 8 and 9).

Figure 9 Number of 09R arrival noise events by aircraft type



33. Figure 10 compares the average measured arrival L_{ASmax} noise levels by aircraft type. Error bars representing one Standard Deviation are also shown to illustrate the variation of measured levels around each average value. Results for the A321, B737, B737MAX and B747 have not been included since the sample sizes are too small to draw any meaningful conclusions.

Figure 10 Average measured 09R arrival L_{ASmax} levels by aircraft type



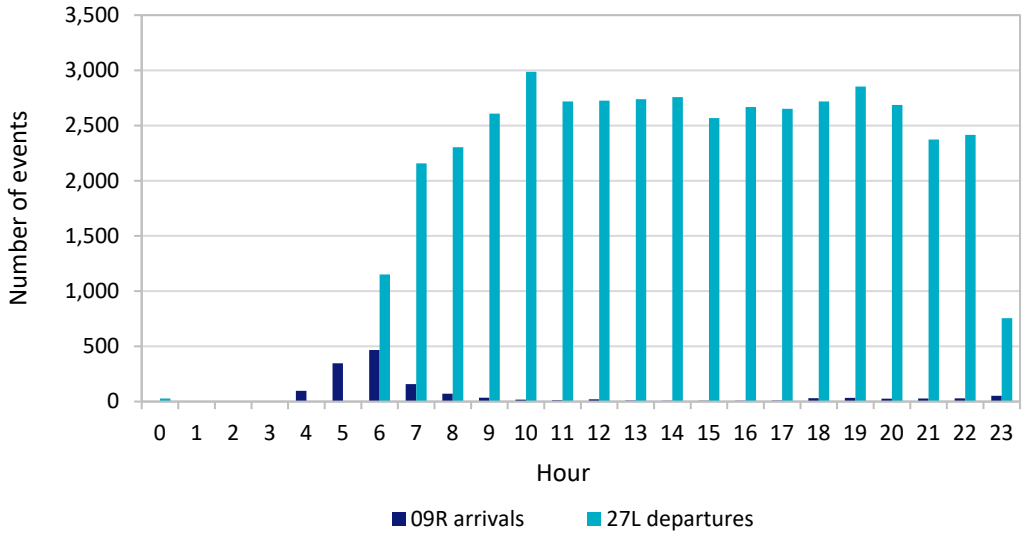
34. The average measured arrival L_{ASmax} across all aircraft types was 89.2 dB. The noisiest aircraft on arrival was the B767, followed by the A380, with average L_{ASmax} levels of 92.1 dB and 91.6 dB, respectively. The quietest aircraft on arrival was the A220, followed by the A320neo (showing the same trend as for the departures), with average L_{ASmax} levels of 85.4 dB and 85.8 dB, respectively. The average arrival

noise level for the re-engined A320neo was approximately 1 dB quieter relative to its older A320 counterpart.

Noise events by hour of the day

35. Figure 11 compares the total number of valid 27L departure noise events and 09R arrival noise events by hour of the day over the entire monitoring period. The graph succinctly illustrates that departures accounted for the vast majority of noise events recorded at Stanwell Moor, again, which is unsurprising given the location of the monitoring site and the overall tendency for westerly operations.

Figure 11 Count of noise events by hour of the day

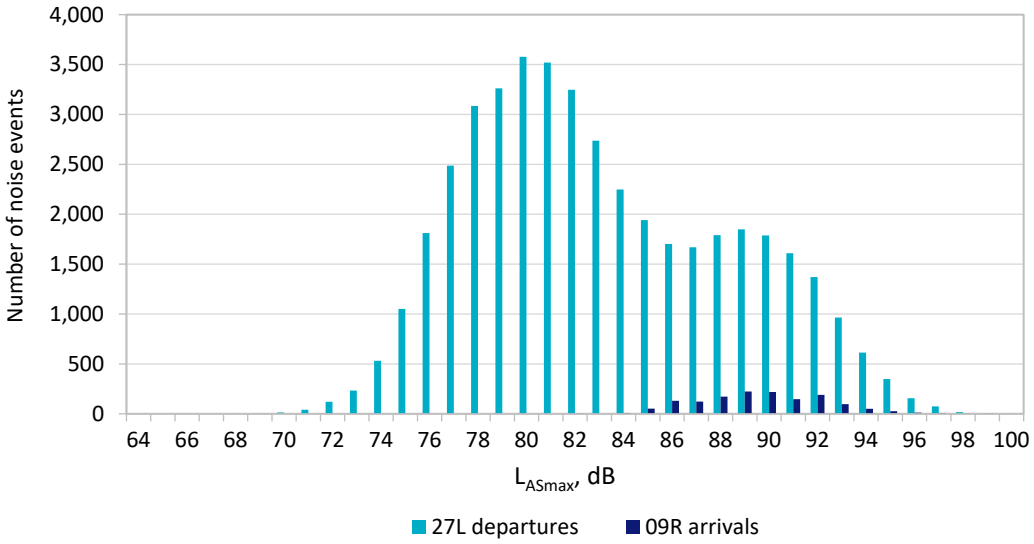


36. The results in Figure 11 also show that the hourly periods with the highest numbers of departure events were between 09:00 and 21:00, which coincides with the busiest period of the day for departures in general. Similarly, noise events from arrivals were most frequent between 06:00 and 07:00, which is typically the busiest hour for 09R arrivals.

Distribution of L_{ASmax} noise levels

37. Figure 12 presents the frequency distribution of L_{ASmax} noise levels measured for 27L departures over the entire monitoring period, with L_{ASmax} levels grouped in 1 dB intervals. Also shown for comparison is the distribution of L_{ASmax} noise levels recorded for 09R arrivals over the same period.

Figure 12 Distribution of L_{ASmax} noise levels for 27L departures and 09R arrivals

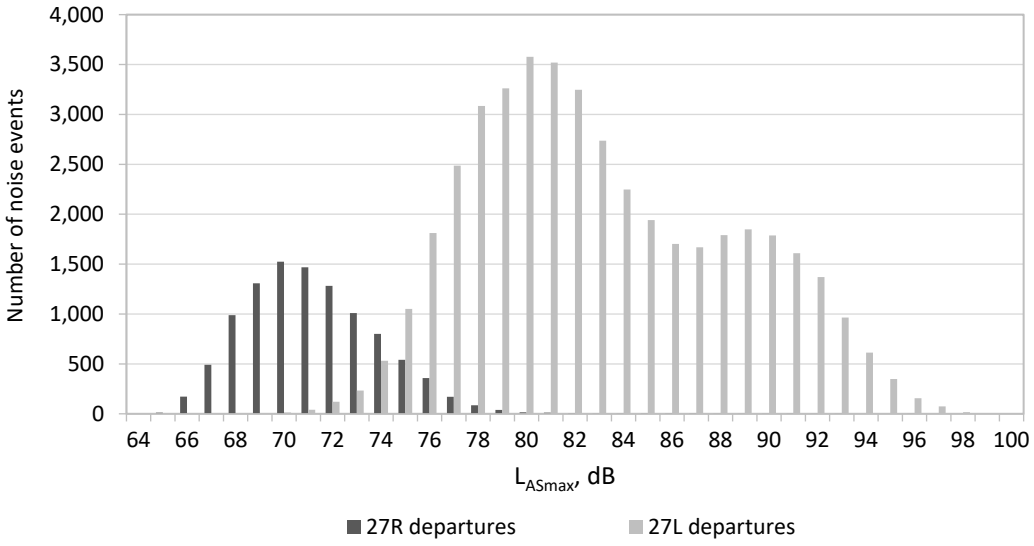


38. Figure 12 illustrates the generally more variable nature of departure noise at the monitor location, evidenced by the fact that the departure distribution exhibits a much wider range of L_{ASmax} values compared to the arrival distribution. Two distinct peaks are also visible in the departure distribution, with the lower peak containing mainly smaller narrow-body aircraft and the higher peak containing mainly larger and noisier wide-body aircraft.[6]

39. Finally, Figure 13 presents the separate L_{ASmax} distributions for departures on the northern runway (27R) and southern runway (27L). The graph illustrates the relatively large difference in measured noise level at the monitor location depending on the runway in use, with the noise values for 27L departures appearing further to the right on the horizontal axis compared to 27R. Since the number of departures from each runway during periods of westerly operations is usually quite similar, Figure 13 also highlights how the 64 dB monitor threshold resulted in a significant proportion of quieter 27R departure noise events remaining undetected.

[6] The average heights of narrow-body and wide-body departures at the monitor location were approximately 1,500 ft and 1000 ft, respectively.

Figure 13 Distribution of L_{ASmax} noise levels for 27R and 27L departures



APPENDIX A

Glossary of terms and abbreviations

A-weighting	A frequency weighting that is applied to the electrical signal within a noise measuring instrument as a way of simulating the way the human ear responds to a range of acoustic frequencies.
dB (or dBA)	Decibel units describing sound level or changes of sound level. It is used in this report to define levels measured on the A-weighted scale.
$L_{A90,T}$	L_{A90} is the A-weighted noise level exceeded for 90% of the measurement time (T) and is commonly used to quantify the level of background noise.
L_{ASmax}	The maximum sound level measured during an aircraft event, using frequency weighting A and time weighting S. Often abbreviated to L_{Amax} or L_{max}.
NPR	Noise Preferential Route. The preferred route for aircraft to fly in order to minimise their noise profile on the ground in the immediate vicinity of the airport.
NTK	Noise and Track Keeping monitoring system. The NTK system associates air traffic control radar data with related data from both fixed (permanent) and mobile noise monitors at prescribed positions on the ground.
Time weighting	Time weighting is used to describe how quickly a sound level meter reacts to changes in sound pressure. There are two commonly used time weightings for environmental noise measurement, Fast (F) and Slow (S). For aviation noise it is standard practice to use Slow weighting.